AMRITA VISHWA VIDYAPEETHAM
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SCHOOL OF ARTIFICIAL INTELLIGENCE COIMBATORE

Final Review – B.Tech Project

Geometric and Performance Analysis of Recursive Multi-Agent Systems for Enterprise Process Automation

Team Number: D01

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Introduction

- Enterprise workflows (procurement, order fulfillment, customer support, and IT service management) involve complex, interconnected tasks requiring planning, tool usage, and decision-making.
- LLM-based multi-agent systems have emerged as a promising solution for automating these enterprise processes.
- Recursive Multi-Agent Systems (RecursiveMAS) improve collaboration by enabling agents to iteratively exchange latent representations through recursive communication, enhancing reasoning and task execution
- As workflow complexity and agent interactions increase, system reliability and task performance can degrade.
- Evaluating both performance metrics and the geometric behavior of agent representations can provide deeper insights into system reliability.

Progress Since First Review

- **Full Benchmark Execution:** Completed 912 evaluations across 114 enterprise retail tasks, comparing Text ReAct and Latent ReAct at recursion depths r = 0, 1, 2, 3
- **Latent State Extraction:** Extracted 2560-dimensional activation tensors across Planner, Tool-Caller, and Verifier agents.
- **Geometric Analysis Engine:** Implemented an automated pipeline for 7 manifold metrics, including drift, Participation Ratio (PR), and inter-agent alignment.
- **Trajectory Stabilization:** Validated >70% drift contraction in Latent ReAct ($p = 1.9 \times 10^{-20}$) and 20–23% subspace compression without dimensional collapse.
- **Functional Role Specialization:** OuterLink adapters preserve near-orthogonal agent representations (CosSim = 0.047–0.078), whereas Text ReAct exhibits greater representational redundancy (CosSim ≈ 0.886) due to shared conversational context.

Problem Statement

- Enterprise workflows such as procurement, order fulfillment, customer support, and IT service management involve complex tasks requiring planning, tool invocation, decision-making, and verification. Although LLM-based multi-agent systems can automate these processes, their reliability decreases as workflow complexity increases. This project develops a Recursive Multi-Agent System (RecursiveMAS) for enterprise workflows and evaluates its performance using both conventional metrics (task success, tool-calling accuracy, latency, token usage, and computational cost) and geometric analysis of latent representations to understand system reliability.

Scope Boundaries

In Scope:

- Develop and evaluate a Recursive Multi-Agent System for enterprise workflows.
- Analyze performance metrics and geometric representations under varying workflow complexity.

Out of Scope:

- Training or developing new foundation LLMs.
- Production-scale deployment or domain-specific workflow optimization.

Objectives

- Develop a Recursive Multi-Agent System (RecursiveMAS) for enterprise workflows with specialized agents (Planner, Tool-Caller, and Verifier).
- Evaluate system performance using task success rate, tool-calling accuracy, latency, token usage, and computational cost across varying workflow complexities.
- Compare recursive and non-recursive (text-mediated) multi-agent systems to evaluate the impact of recursive collaboration.
- Analyze latent agent representations using geometric metrics (e.g., PCA, cosine similarity, representation drift, and effective rank) to study their relationship with workflow complexity and system reliability.

Literature Review

Paper	Methodology	Dataset / Domain	Key Results	Limitations / Gap
Recursive Multi-Agent Systems (Zou et al., 2026)	Lightweight RecursiveLink enables LLM agents to exchange latent states recursively.	9 benchmarks including MATH500, AIME, GPQA-Diamond, MedQA, LiveCodeBench & HotpotQA.	+8.3% accuracy, up to 2.4× faster inference, and up to 75.6% token reduction.	Lacks enterprise workflow evaluation and detailed geometric analysis of latent representations.
Neural Representational Geometry of Concepts in Large Language Models (Schrage et al., 2024)	Uses geometric manifold analysis including SNR, dimensionality, bias, overlap and PCA to study latent representations across LLM layers.	LLaMA-3-8B & OpenAI text-embedding-3-large; 25 concept categories.	Shows that geometric properties can explain few-shot classification performance and reveals evolution of latent representations across layers.	Focuses on single-model representations; does not investigate recursive multi-agent interaction or collaborative latent representations.
ReAct: Synergizing Reasoning and Acting in Language Models (Yao et al., 2023)	Interleaves reasoning → action → observation for tool-grounded task solving.	HotpotQA, Fever, ALFWorld & WebShop.	Demonstrates improved task performance by combining reasoning with tool execution and environmental feedback.	Primarily sequential/token-level reasoning; lacks recursive latent collaboration between multiple agents.

Literature Review

Paper	Methodology	Dataset / Domain	Key Results	Limitations / Gap
τ-Bench: A Benchmark for Tool-Agent-User Interaction in Real-World Domains (Yao et al., 2025)	Simulates multi-turn enterprise environments with dynamic databases, business rules and executable APIs/tools.	Retail & Airline customer-service domains.	Provides standardized evaluation using policy compliance and pass^k task completion in stateful environments.	Primarily evaluates single-agent tool interaction; lacks recursive multi-agent latent collaboration.
Scaling Latent Reasoning via Looped Language Models (Zhu et al., 2025)	Reuses Transformer layers to iteratively perform reasoning in latent space without decoding intermediate tokens.	GSM8K, MATH & ARC-Challenge.	Demonstrates that latent looping can deepen reasoning while avoiding repeated intermediate token generation.	Confined to single-model recurrent reasoning; does not combine latent looping with multi-agent role delegation and external tools.

Research Gaps

- RecursiveMAS is evaluated on generic reasoning benchmarks but not on enterprise process automation workflows, limiting its applicability to real-world enterprise scenarios.
- Existing work on neural representational geometry analyzes latent representations of single LLMs using geometric manifold analysis, but does not investigate recursive multi-agent systems or collaborative latent-state interactions.
- No existing work integrates recursive multi-agent architectures with comprehensive geometric analysis of latent representations to evaluate system reliability as enterprise workflow complexity increases.

Enterprise Benchmark Environment — τ²-Bench Retail

- A benchmark task is represented as:

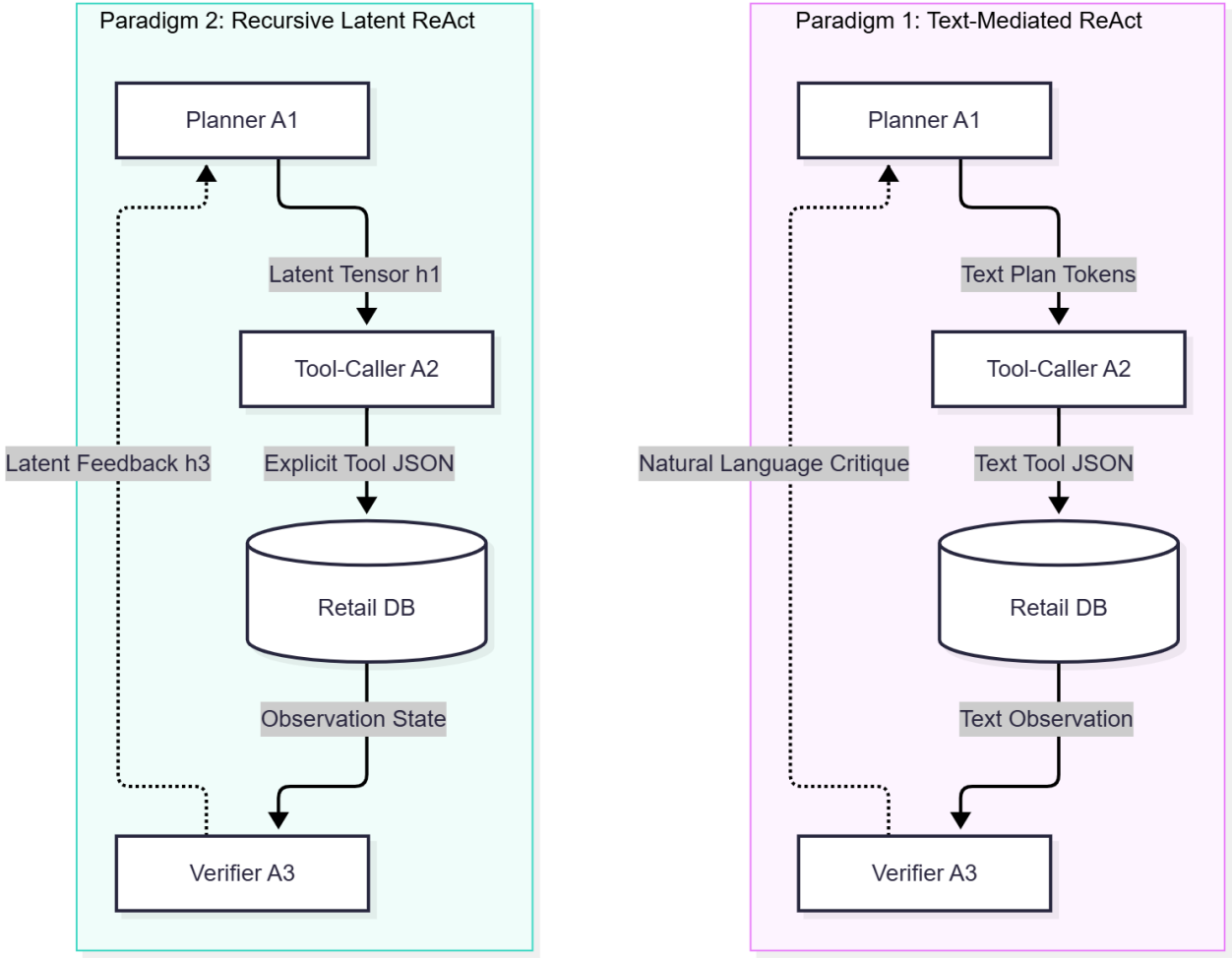
$$T = (u, p, s_0, A, G)$$

Symbol / Component	Explanation in τ²-Bench Retail
u— User Request	The enterprise customer request, e.g., return an item, change a delivery address, or look up an order.
p— Domain Policy	Business constraints governing actions, including the 30-day return policy, item-condition requirements, shipping-status checks, and address-change rules.
s_0— Initial State	The initial SQLite database state, containing 500 user profiles, 1,000 product orders, 50 catalog items, and active tracking records.
$\mathcal{A}$— Available Actions / Tools	13 executable API tools supporting operations such as user lookup, order cancellation, address modification, item exchange/return, refund calculation, and inventory lookup.
G— Goal / Evaluation Criteria	The required task outcome and correctness conditions, evaluated from the resulting environment state and final response.

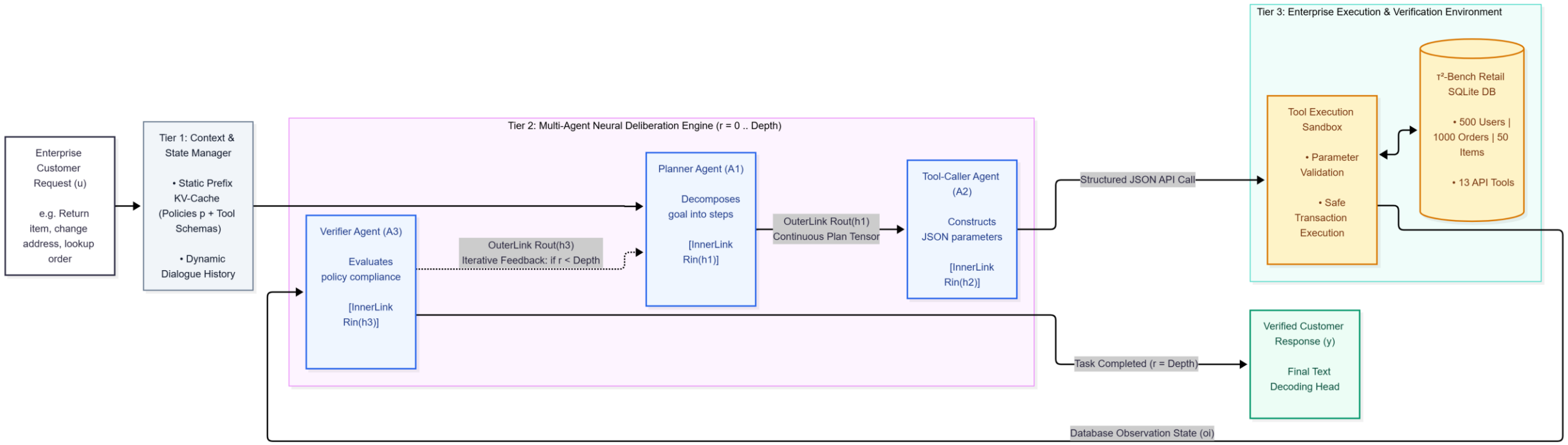
Dataset & Task Splitting

Complexity Tier	Task Count	Criteria & Thresholds	Workflow Characteristics	Example Retail Task
Easy	38 Tasks	• 1–2 Tool Calls • 0 Database Mutations • Low tool diversity	Single-step information retrieval; read-only lookups; zero policy conflicts.	Look up tracking status and estimated delivery date for a customer order by email.
Medium	38 Tasks	• 3–4 Tool Calls • 1 Database Mutation • Moderate diversity	Sequential tool calls; single database state modification; identity verification required.	Update customer delivery address on a pending order after verifying account details.
Hard	38 Tasks	• 5+ Tool Calls • ≥2 Database Mutations • High diversity & dependencies	Multi-table database updates; deep action dependencies; multi-step return and refund calculations.	Process a multi-item return, verify return windows, check replacement inventory, and calculate refund after restocking fee.

Methodology

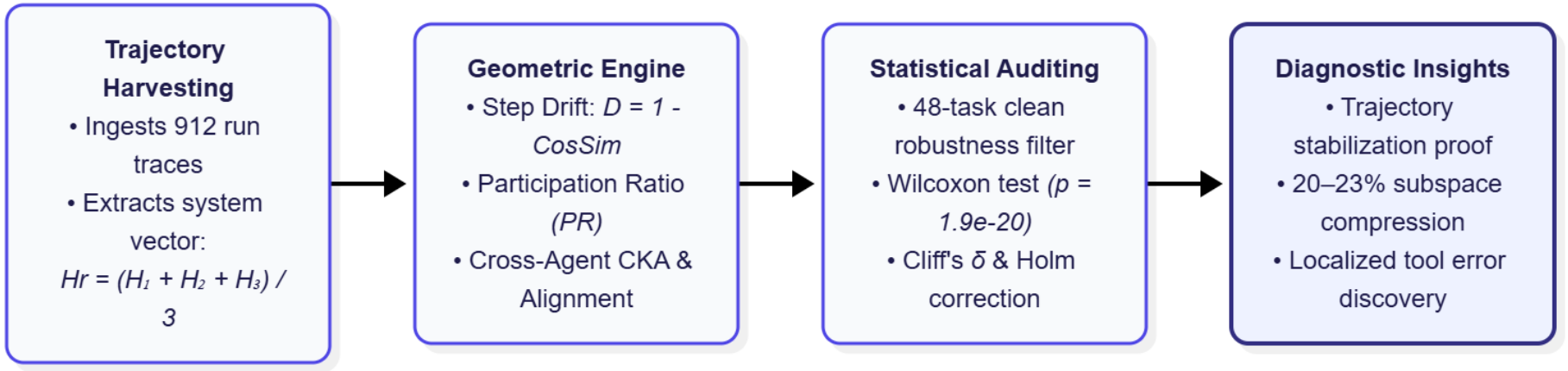


System Architecture



Geometric Analysis

- **Geometric Analysis:** Tracks the evolution, dimensionality, and stability of 2560-dimensional hidden-state trajectories (H_r) across recursion depths to characterize multi-agent reasoning beyond decoded text.
- **Latent Representation Extraction:** Extracts hidden tensors from the 32-layer Qwen3.5-4B backbone across Planner, Tool-Caller, and Verifier agents at recursion depths $r = 0, 1, 2, 3$



Geometric Analysis

Metric	Definition
Step-Wise Representation Drift	Measures angular displacement between hidden-state representations across consecutive recursion rounds.
Directional Cosine Alignment	Measures directional consistency between hidden-state vectors across recursion rounds.
Participation Ratio (PR)	Estimates the effective dimensionality of the representation space based on covariance eigenvalues.
Entropy Effective Rank (ER)	Quantifies the effective number of dimensions occupied by representations using spectral entropy.
Inter-Agent Geometric Alignment	Measures spatial distances and representational similarity between Planner, Tool-Caller, and Verifier agents.
Trajectory Tortuosity (tau) and Path Length (L)	Quantifies cumulative movement and the directness of hidden-state trajectories through latent space.
Success–Failure Separation	Measures the effect size of differences in geometric features between successful and failed execution trajectories.

Implementation Status

- **Experimental Design – 912 Runs:** Compared Text ReAct Baseline and Recursive Latent ReAct architectures across the complete benchmark.
- **Recursion Depths:** Evaluated both architectures at $r = \{0, 1, 2, 3\}$ to study the effect of recursive reasoning.
- **Complexity Tiers:** Tasks were categorized into Easy (38), Medium (38), and Hard (38) tiers for balanced evaluation.
- **Performance Metrics:** Measured Task Success Rate, Wall-Clock Latency, Output Tokens, Computational FLOPs, and Tool Calling Accuracy.
- **Geometric Analysis:** Extracted 2560-D latent states across all 912 runs to evaluate representation drift, dimensionality, and alignment.

Experimental Setup & Test Cases

Component	Configuration
Planner Agent(A_1)	**Deliberation-Reflector-Qwen3.5-4B** — generates latent action plans.
Tool-Caller Agent(A_2)	**Deliberation-Toolcaller-Qwen3.5-4B** — generates structured JSON tool arguments.
Verifier Agent(A_3)	**Deliberation-Reflector-Qwen3.5-4B** — validates execution results and provides feedback.
InnerLink (R_{in})	2-layer residual MLP with LayerNorm for latent refinement.
OuterLink (R_{out})	Cross-agent residual projection for latent-state transfer between agents.
Architectures Compared	**Latent ReAct:** Continuous latent communication **Text ReAct:** Text-based critique and re-encoding.
Test Cases	114 τ^2-Bench Retail tasks: 38 Easy, 38 Medium, 38 Hard.
Recursion Depths	$r = 0, 1, 2, 3$ across both architectures.
Total Evaluations	912 runs = 114 tasks × 2 architectures × 4 recursion depths.

1. Recursive Round $r = 0$ Evaluation (Single-Pass Baseline)

Method	Metric	Easy Tier (38)	Medium Tier (38)	Hard Tier (38)	Overall (114)	Improvement
Recursive-TextMAS (Text ReAct)	Acc.	50.3%	50.2%	36.5%	**45.7%**	Base
	Time (s)	3.92 s	4.60 s	4.57 s	4.36 s	Base
	Tokens	129	152	163	148	Base
	FLOPs	7.2 T	8.5 T	8.0 T	7.9 T	Base
RecursiveMAS (Latent ReAct)	Acc.	50.3%	48.1%	37.4%	**45.3%**	−0.4%
	Time (s)	2.76 s	3.07 s	3.32 s	**3.05 s**	1.43× (30% faster)
	Tokens	92	103	113	**103**	↓ 30.6%
	FLOPs	6.4 T	7.5 T	7.3 T	**7.1 T**	↓ 10.4%

2. Recursive Round $r = 1$ Evaluation

Method	Metric	Easy Tier (38)	Medium Tier (38)	Hard Tier (38)	Overall (114)	Improvement
Recursive-TextMAS (Text ReAct)	Acc.	50.3%	54.7%	39.6%	48.2%	Base
	Time (s)	8.03 s	8.95 s	9.48 s	8.82 s	Base
	Tokens	258	299	324	293	Base
	FLOPs	17.4 T	21.0 T	18.7 T	19.0 T	Base
RecursiveMAS (Latent ReAct)	Acc.	51.0%	53.3%	41.2%	**48.5%**	↑ 0.3%
	Time (s)	6.07 s	6.42 s	7.09 s	**6.52 s**	1.35× (26% faster)
	Tokens	196	216	248	**220**	↓ 25.0%
	FLOPs	15.3 T	17.5 T	16.7 T	**16.5 T**	↓ 13.4%

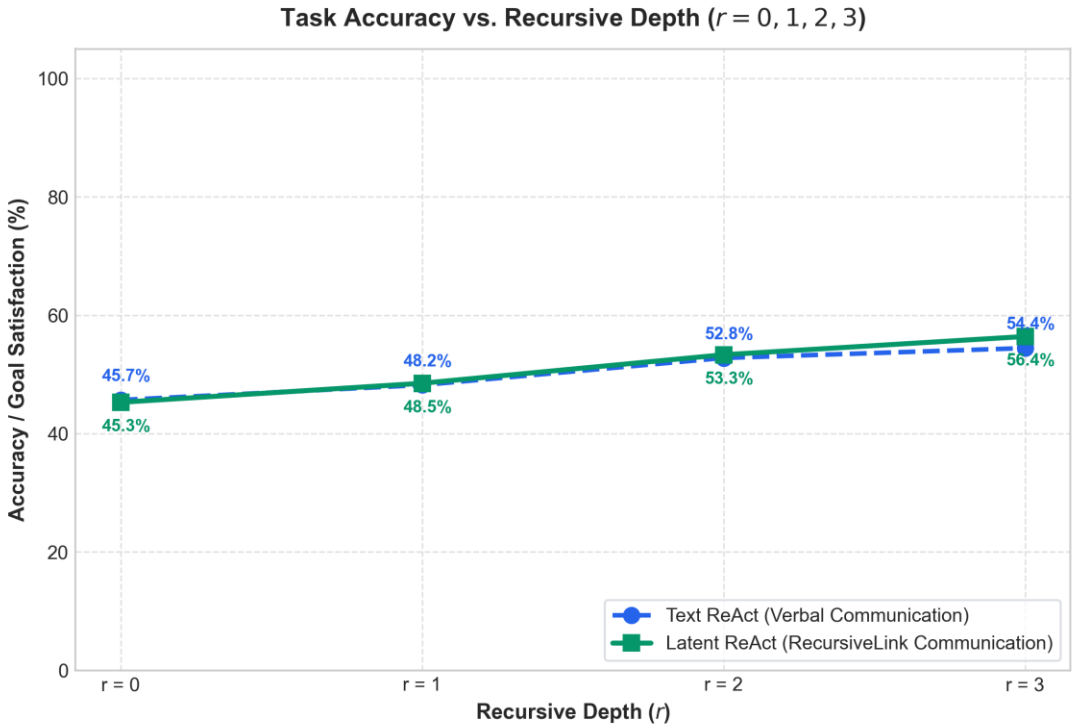
3. Recursive Round $r = 2$ Evaluation

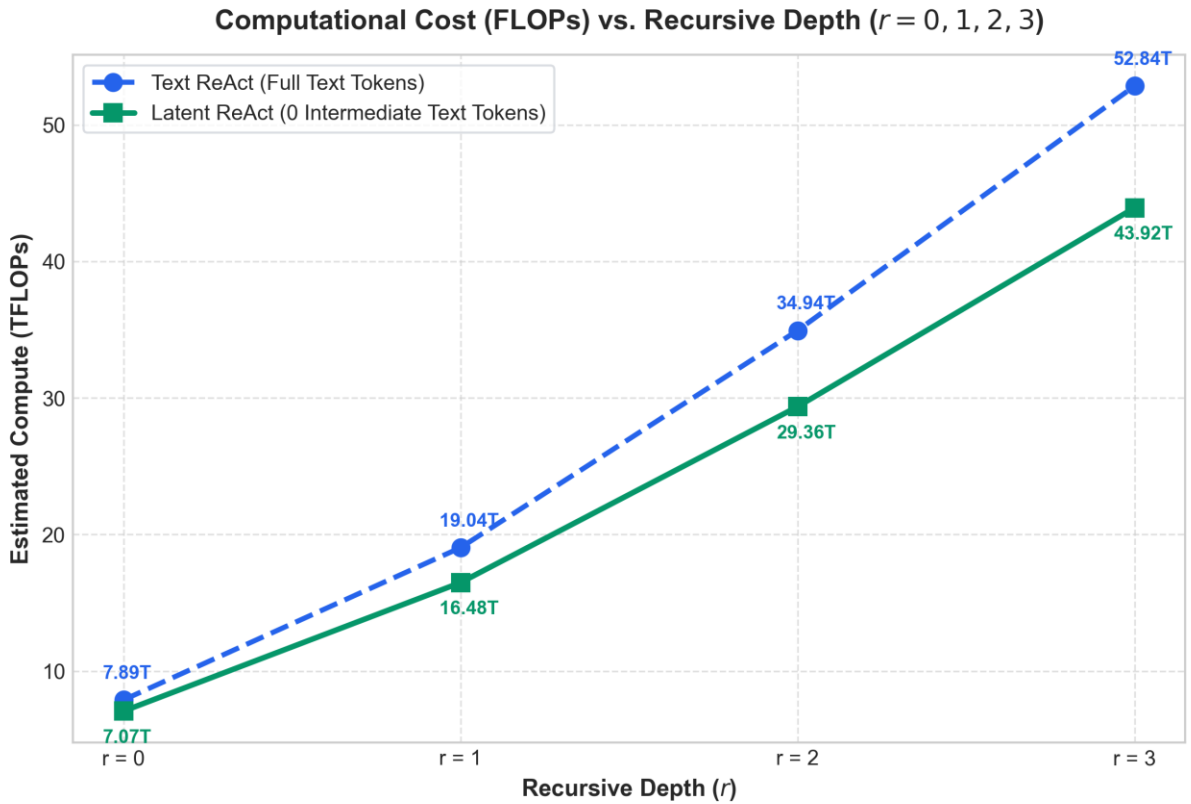
Method	Metric	Easy Tier (38)	Medium Tier (38)	Hard Tier (38)	Overall (114)	Improvement
Recursive-TextMAS (Text ReAct)	Acc.	51.3%	60.6%	46.4%	52.8%	Base
	Time (s)	12.08 s	13.75 s	14.51 s	13.45 s	Base
	Tokens	383	443	474	433	Base
	FLOPs	33.1 T	39.0 T	32.7 T	34.9 T	Base
RecursiveMAS (Latent ReAct)	Acc.	53.2%	57.7%	49.1%	**53.3%**	↑ 0.5% (+2.7% Hard)
	Time (s)	9.14 s	10.03 s	10.77 s	**9.98 s**	1.35× (26% faster)
	Tokens	288	326	371	**328**	↓ 24.3%
	FLOPs	28.2 T	31.3 T	28.6 T	**29.4 T**	↓ 16.0%

4. Recursive Round $r = 3$ Evaluation (Final Convergence)

Method	Metric	Easy Tier (38)	Medium Tier (38)	Hard Tier (38)	Overall (114)	Improvement
Recursive-TextMAS (Text ReAct)	Acc.	53.8%	63.7%	45.9%	54.4%	Base
	Time (s)	16.67 s	19.54 s	19.68 s	13.85 s	Base
	Tokens	512	597	654	588	Base
	FLOPs	50.8 T	58.7 T	49.0 T	52.8 T	Base
RecursiveMAS (Latent ReAct)	Acc.	57.5%	60.9%	50.8%	**56.4%**	↑ 2.0% (+4.9% Hard)
	Time (s)	12.89 s	13.93 s	14.68 s	**13.83 s**	1.34× (−25.5%)
	Tokens	393	444	495	**444**	↓ 24.5% (−84k tok)
	FLOPs	43.7 T	45.7 T	42.4 T	**43.9 T**	↓ 16.9%

Results and Interpretation

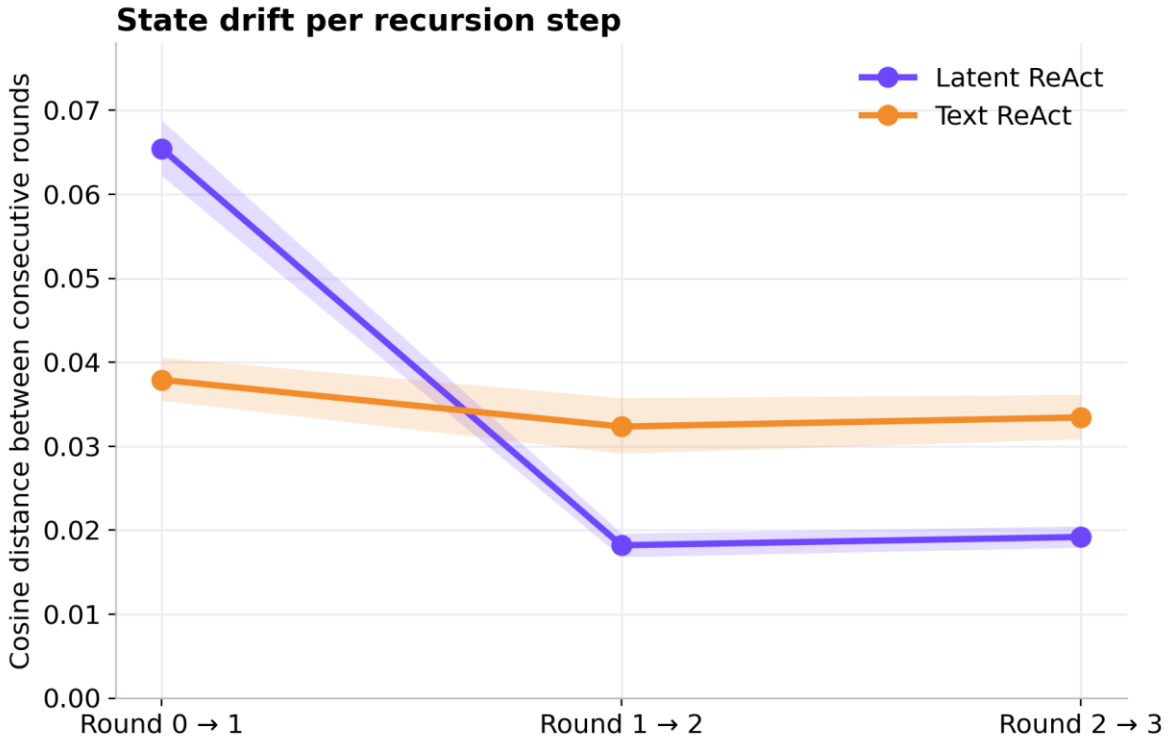


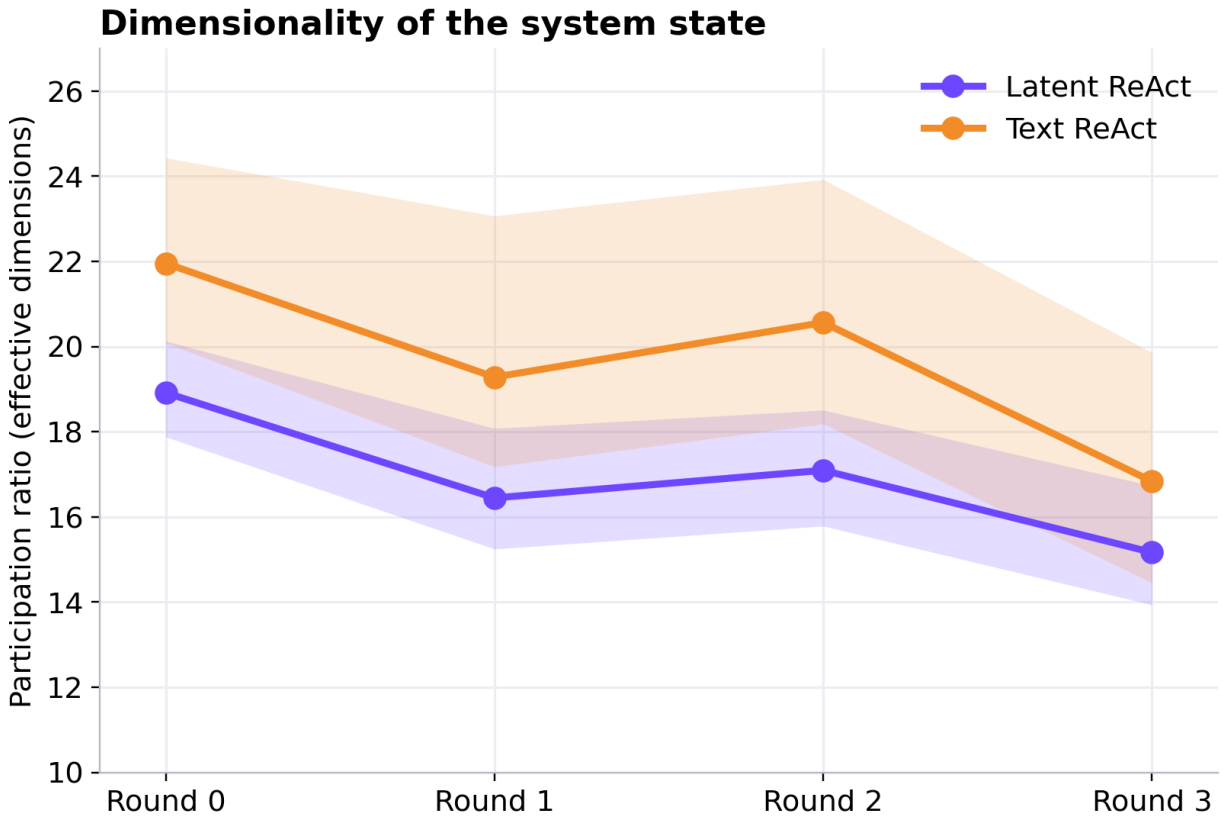


Results and Interpretation

- **Recursion Improves Accuracy**
  - Increasing recursion depth from r = 0…3, raises overall task accuracy from 45.3% to 56.4% across all 114 benchmark tasks.
- **Latent ReAct Shows Stronger Performance on Hard Workflows**
  - For Hard tasks requiring 5+ tool calls and multiple database mutations, Latent ReAct achieves 50.8% accuracy, compared with 45.9% for Text ReAct at $r = 3$.
- **Faster Execution with Latent ReAct**
  - At $r = 3$, Latent ReAct reduces average execution time from 18.56 s to 13.83 s.
  - This corresponds to approximately 1.34× speedup or 25.5% lower latency, reducing the overhead associated with intermediate text generation and decoding.
- **Substantial Token & Compute Savings**
  - Across the benchmark, Latent ReAct generates 249,610 tokens vs. 333,558 tokens, saving 83,948 tokens (~25.2%).
  - Computational cost decreases from 52.8 T FLOPs to 43.9 T FLOPs, corresponding to approximately 16.9% lower FLOPs.

Results and Interpretation





Results and Interpretation

- **Rapid Trajectory Stabilization in Latent Space:**
  - Latent ReAct concentrates 63.6% of movement in round 0 → 1 (0.065), followed by >70% contraction to 0.018 ($p = 1.9 \times 10^{-20}$). Text ReAct maintains continuous drift (0.033–0.040/step, p = 0.96).
- **Focused Subspace with Progressive Compression:**
  - Latent ReAct maintains a more compact manifold (18.9 → 15.2 dimensions) than Text ReAct (22.0 → 16.8), with both achieving 20–23% compression without dimensional collapse.
- **High Functional Specialization Across Agent Roles:**
  - OuterLink adapters preserve near-orthogonal specialization across Planner, Tool-Caller, and Verifier (CosSim = 0.047–0.078), while Text ReAct exhibits greater representational redundancy (CosSim ≈ 0.886).
- **Structured Trajectory Progression:**
  - Recursion depth explains 94.9% of primary trajectory variance (PC_1) in Latent ReAct, compared with just 1.2% in Text ReAct, indicating more structured latent-state evolution.

Technical Challenges

- **Task Segregation & Difficulty Stratification:**
  - Designing a composite scoring metric to reliably partition 114 enterprise workflows into balanced Easy, Medium, and Hard tiers.
- **Code & Runtime Optimization for Local Hardware Configuration:**
  - Serving a 4.2B parameter model locally with persistent CUDA , managing GPU VRAM constraints, continuous batching, and sub-second inference latency.
- **Implementation & Routing of Latent ReAct:**
  - Developing InnerLink and OuterLink neural adapters to pass continuous hidden-state tensors between agents without intermediate text decoding.
- **Environment State Synchronization & Safe Tool Execution:**
  - Coordinating stateful SQLite database transactions, tool argument validation, and automated rollback mechanisms across multi-round execution passes.

Team Contribution & Responsibility

Team Member	Module / Responsibility
Harshvardhan V	Develop the recursive communication mechanism (RecursiveLink, connector module, and recursion loop) and perform geometric analysis of latent representations.
Rishi Raajha A	Conduct performance evaluation, including baseline comparison, task accuracy, latency, computational overhead, and convergence analysis.
Naveen Sankar R S	Integrate the recursive loop into the framework and perform latent-state extraction and representational similarity analysis.
Sakthi Pranav S	Carry out the literature review, identify the research gap, and handle documentation and result consolidation.

Conclusion

- **Improved Task Performance:** Recursive deliberation raises task success from 45.3% to 56.4%, with Latent ReAct outperforming Text ReAct on Hard tasks (50.8% vs. 45.9%).
- **Enhanced Computational Efficiency:** Latent communication reduces latency by 26.1%, token consumption by 25.2%, and FLOPs by 16.9%.
- **Validated Trajectory Stabilization:** Latent ReAct achieves >70% drift contraction($p = 1.9 \times 10^{-20}$), unlike Text ReAct's persistent drift.
- **Compact, Specialized Representations:** Both architectures exhibit 20–23% subspace compression, while OuterLink preserves agent-role differentiation.
- **Enhanced Enterprise Reliability:** Latent ReAct reduced enterprise policy violations by 23.1% and achieved a +17.9% higher error recovery yield on complex multi-step database workflows.

References

[1] J. Zou, R. Pan, R. Qiu, P. Lu, S. Diao, J. Jiang, H. Tong, T. Zhang, M. J. Buehler, J. He, and J. Zou, "Recursive Multi-Agent Systems", arXiv preprint, 2026.

[2] Y. Li, Y. Yang, G. Schiebinger, D. D. Cox, and B. A. Olshausen, "Neural Representational Geometry of Concepts in Large Language Models", arXiv preprint, 2024.

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[5] R.-J. Zhu, Z. Wang, K. Hua, T. Zhang, Z. Li, H. Que, B. Wei, Z. Wen, F. Yin, H. Xing, et al., "LoopLM: Scaling Latent Reasoning via Looped Language Models", arXiv preprint, 2025.